

RATNESH KR. JOSHI

Ph.D. Scholar

Building transparent and trustworthy conversational AI

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EDUCATION

2019 – 2026 Ph.D., Computer Science and Engineering, Indian Institute of Technology Patna
Thesis: Investigating Explainability Aided Conversational Systems for Imparting Transparency, Faithfulness, and Trust. *CPI:* 8.3

2015 – 2017 M.Tech., Computer Science and Engineering, Central University of Rajasthan
Thesis: Long Short-Term Memory and Hidden Markov Models Hybrid Modeling.

2010 – 2014 B.Tech., Computer Science and Engineering, College of Engineering Roorkee

EXPERIENCE

Mar 26 – Now Research Scientist, Siemens Technology, Bangalore, India

- Focused research in **AI for industrial applications**.

Mar 26 – Now Post-Doc Researcher, IIIT Bangalore, Bangalore, India

- Focused research in **Multimodality and Vision-Language Models (VLMs)**.

Mar – Sep 2025 Internship on AI System Design, Samsung Research Institute, Noida, India

- **Architected** and implemented a Conversational AI system using Retrieval Augmented Generation (RAG), ensuring **98%** contextually accurate responses across complex text and code repositories.
- **Validated** system performance, achieving a high **FactTool score of 0.87** for factual consistency and reduced hallucinations.

Apr – Oct 2023 Internship on Behavioral Influencing, Technology Innovation Lab - Accenture, Bengaluru

- **Contributed** in the engineering of a conversational agent designed to promote environmental sustainability through behavioral influencing techniques.
- **Leveraged** advanced LLMs and ethical negotiation tactics (rational appeal, inspirational appeals, factual statistics, associations) to successfully guide user in **approx. 60%** of interactions.
- **Analyzed** over **200** interaction logs to refine persuasion strategies, focusing on factual grounding and user associations.

PROJECTS

Sevak - An Intelligent Indian Language Chatbot

- **Orchestrated** a team of collaborators to build pluggable chatbot for Railways, Judiciary, and Healthcare domains.
- **Supervised** the end-to-end data pipeline, including collection and cleaning, to deploy robust English and Hinglish (code-mixed) models.
- **Devised** the system integration strategy and executed server deployment, successfully connecting the backend to multiple front-end interfaces like WhatsApp, FB Messenger, proprietary web interface.

Attribution-Grounded Generation

- **Developed** novel methods to infuse saliency and feature attributions into dialogue generation, enhancing the faithfulness and transparency of task-oriented responses.
- **Established** a saliency-guided framework to detect and mitigate inherent biases in Large Language Models, achieving a **50% reduction** in incorrectly gendered responses.

RAG Based Conversational Assistant

- **Constructed** a proprietary Conversational AI pipeline utilizing Retrieval Augmented Generation (RAG).
- **Empowered** the system to synthesize internal, proprietary data, delivering grounded responses over text and code with **significantly reduced hallucination rates**.

SKILLS

Programming & Frameworks

Python

PyTorch

Hugging Face

LangChain

Scikit-learn

Pandas

Matplotlib

Google Colab

Core AI/ML Techniques

Machine Learning

Deep Learning

NLP

RL

RAG

Agentic AI

Statistical Modeling

Specialized Expertise

Explainable AI

Conversational AI

Multimodality

Vision-Language Models (VLMs)

Model Debiasing

Professional & Analytical

Research & Publication

Technical Instruction

Project Management

Collaboration & Persuasion

PUBLICATIONS

- **Joshi, R. K.**, Chatterjee, A., & Ekbal, A. "Saliency infused dialogue response generation: Improving task-oriented text generation using feature attribution." *Expert Systems with Applications*, 124283.
- **Joshi, R. K.**, Chatterjee, A., & Ekbal, A. "Saliency Guided Debiasing: Detecting and mitigating biases in LMs using feature attribution." *Neurocomputing*, 563, 126851.
- Priya, P., Desai, Y., **Joshi, R.**, et al. "TRIP NEGOTIATOR: A Travel Persona-aware Reinforced Dialogue Generation Model for Personalized Integrative Negotiation in Tourism." *Findings of EMNLP 2024*.
- **Joshi, R. K.**, Sengupta, S., & Ekbal, A. "From Hallucinations to Facts: Enhancing Language Models with Curated Knowledge Graphs." *arXiv preprint arXiv:2412.18672*.
- **Joshi, R. K.**, Priya, P., Desai, V., et al. "Strategic prompting for conversational tasks: A comparative analysis of large language models..." *arXiv preprint arXiv:2411.17204*.
- **Joshi, R. K.**, Chatterjee, A., & Ekbal, A. "Towards explainable dialogue system: Explaining intent classification using saliency techniques." *Proceedings of ICON*.

ORGANIZATIONAL RESPONSIBILITIES

- **Lecturer & Organizer**, GIAN Workshop on Deep Learning for Conversational AI, IIT Patna (Mar 2025)
- **Webmaster**, 20th International Conference on Natural Language Processing (ICON-2023), Goa University (Dec 2023)
- **Lecturer & Organizer**, GIAN Workshop on Deep Learning for Conversational AI, IIT Patna (Apr 2022)
- **Webmaster**, 16th International Conference on Natural Language Processing (ICON-2020), IIT Patna (Dec 2020)
- **Lecturer & Organizer**, CEP Course on Deep Learning for NLP, IIT Patna (Jan 2020)

CERTIFICATIONS

Google Advanced Data Analytics

Google

 Issued Nov 2025

 Show Credential

Natural Language Processing

DeepLearning.AI

 Issued Nov 2020

 Show Credential

REFERENCES

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